

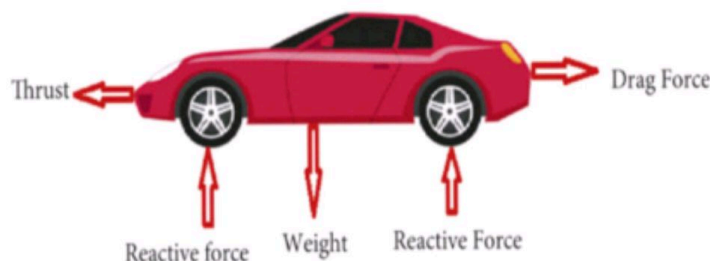
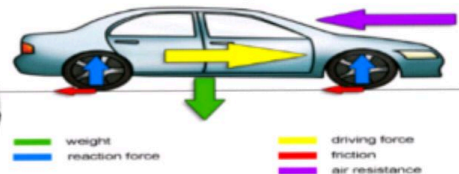


UNIT-1.0

VEHICLE MOVEMENT

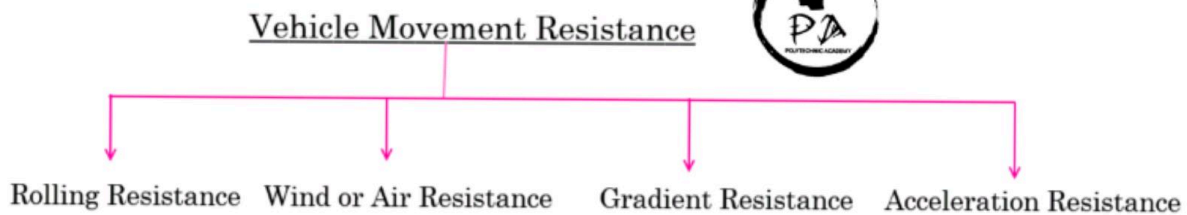
Introduction :-

- The vehicle movement is concerned with the different type of forces which act on a vehicle when it propells.
- Newton's second law of motion played the important role in vehicle movement.
- According to Newton's second law "the acceleration of an object is proportional to the net force exerted on it".
- Hence a vehicle accelerates when the net force acting on it is not zero.
- In a vehicle several forces act on it and the net or resultant force governs the motion according to the Newton's second law.
- The propulsion unit of the vehicle delivers the force necessary to move the vehicle forward.
- The force of the propulsion unit helps the vehicle to overcome the resisting forces due to gravity, air and tire resistance.
- The acceleration of the vehicle depends on :-
 - a) the power delivered by the propulsion unit.
 - b) the road condition
 - c) the aerodynamics of the vehicle
 - d) the composite mass of the vehicle.



Vehicle Movement Resistance :-

- A moving vehicle opposed by various external forces known as resistances.
- A moving vehicle has to overcome the all resistance.
- There are following types of resistance for moving vehicle



1. Rolling Resistance (R_R) :-

- Rolling resistance is friction between tyre and road surface while tyre rolling.
- When a vehicle rolls, it rolls with its tyres in contact with the road surface.
- The relative motion of two hard surfaces produces a friction.

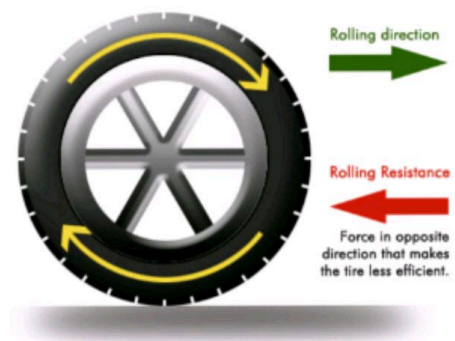
- Rolling resistances depend on the following parameters.

- (1) Road characteristics
- (2) Tyre characteristics
- (3) Weight of the vehicle
- (4) Vehicle speed.

- The value of rolling resistance is determine by

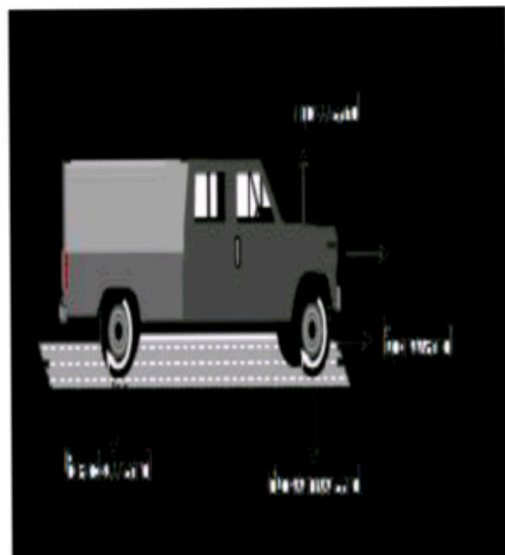
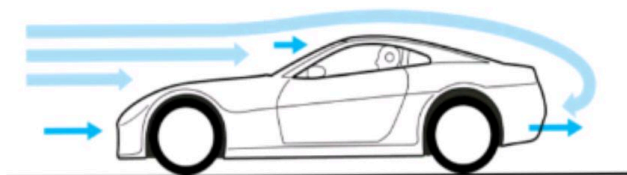
$$(R_R) = K_R \cdot w = K_R \cdot m \cdot g$$

where, K_R = Constant of rolling resistance.
 w = weight of vehicle
 m = mass of vehicle in kg.
 g = Acceleration due to gravity = 9.81 m/s^2 .



2. Wind or Air Resistance :-

- It is the resistance offered by air to the vehicle motion.
- It depends on the following factors.
 - (1) Size of vehicle
 - (2) Shape of vehicle
 - (3) Speed of vehicle, and
 - (4) Wind velocity.



$$\text{Air resistance } R_A = \left(\frac{1}{2} \rho C_d\right) A.V^2$$

ρ = Air density in kg/m^3

C_d = Coefficient of drag

A = Cross sectional area.

V = Vehicle speed

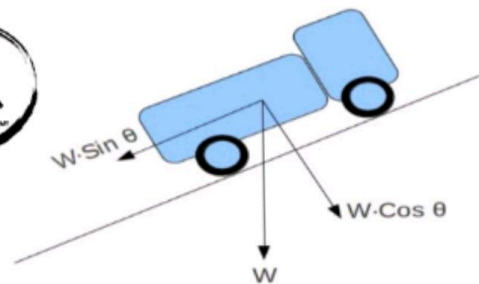


$$K_A = \text{Coefficient of air resistance} = \frac{1}{2} \rho C_d$$

$$\therefore \text{Air resistance } R_A = K_A.A.V^2 \text{ Km/h}$$

3.Gradient Resistance :-

- The gradient resistance is due to slope or steepness of road gradient.
- If the vehicle is trading uphill at a slope of θ , then the weight of the vehicle, W has two components: one perpendicular to the road surface (with a value $W \cos \theta$) and the other along the road surface (with a value $W \sin \theta$).
- The component along the road surface is the one that tries to restrict the motion.
- The gradient resistance is given by :



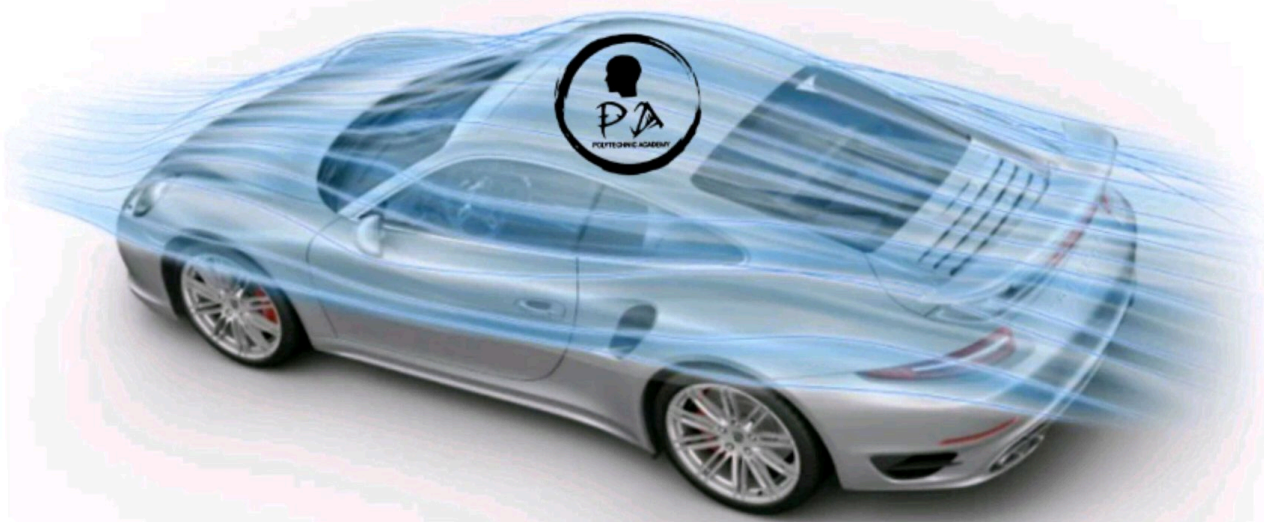
$$R_g = W \sin \theta = m g \sin \theta$$

W = weight of vehical

m = mass of vehicle in kg

g = Acceleration due to gravity = 9.81 m/s^2

Aerodynamic Drag



Aerodynamic Drag

- Aerodynamic drag is a force that the oncoming air applies to a moving car.
- It is the resistance offered by the air to the running car.
- So when a car is moving, it displaces the air.
- However, it affects the car's speed and performance. Drag should be minimum as much as possible.
- Technically, it is the aerodynamic drag or the friction offered by the air to a vehicle.
- This drag force depends mainly on the front area of the vehicle, side mirrors and many other factors.

Equation of Aerodynamic Drag



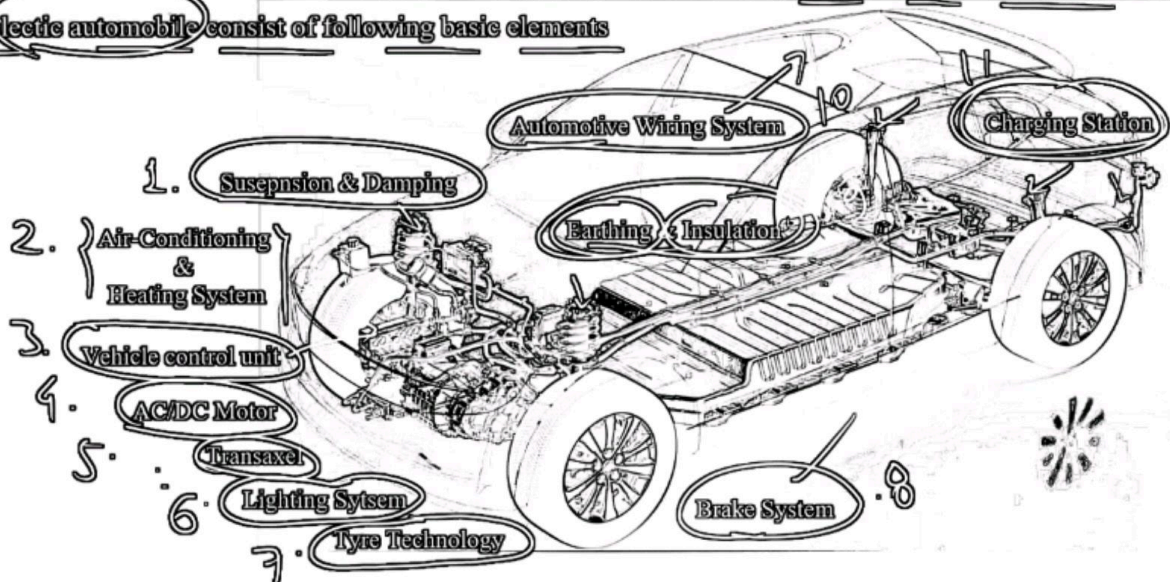
Aerodynamic drag force can be calculated by eqⁿ

$$F_{ad} = \frac{1}{2} \rho A C_d v^2$$

- The drag coefficient depends upon the shape of the object and is always associated with the surface area.
- A lower value of drag coefficient indicated low aerodynamics of the object.
- Nowadays, cars have an average drag coefficient of 0.40 and some cars have as low as 0.26. Such vehicles are said to have the best aerodynamic design.
- Also, the power you need while moving a body through the air increases speed.
- So , above 60 km/h, you need a lot of power to push the car to overcome the aerodynamic drag.
- The reduction in drag can help decrease a vehicle's wind/air resistance. Thus, it helps to increase fuel efficient to some extent.
- The vehicl's frontal area and the drag coefficient are the two main factors that impact the overall drag effect.
- To eliminate the aerodynamic-drag, the manufacturers typically work on the external components.
- These include the roof rack, mud flaps, rear spoiler, side mirrors, radio antenna, windshield wipers etc.



An electric automobile consist of following basic elements



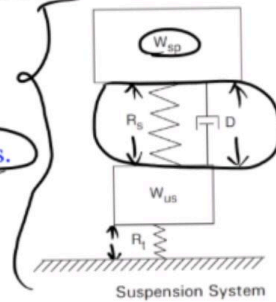
Suspension System

- All the parts which perform the function of isolating the automobile from the road shocks are collectively known as suspension system.

Need of suspension system/Function

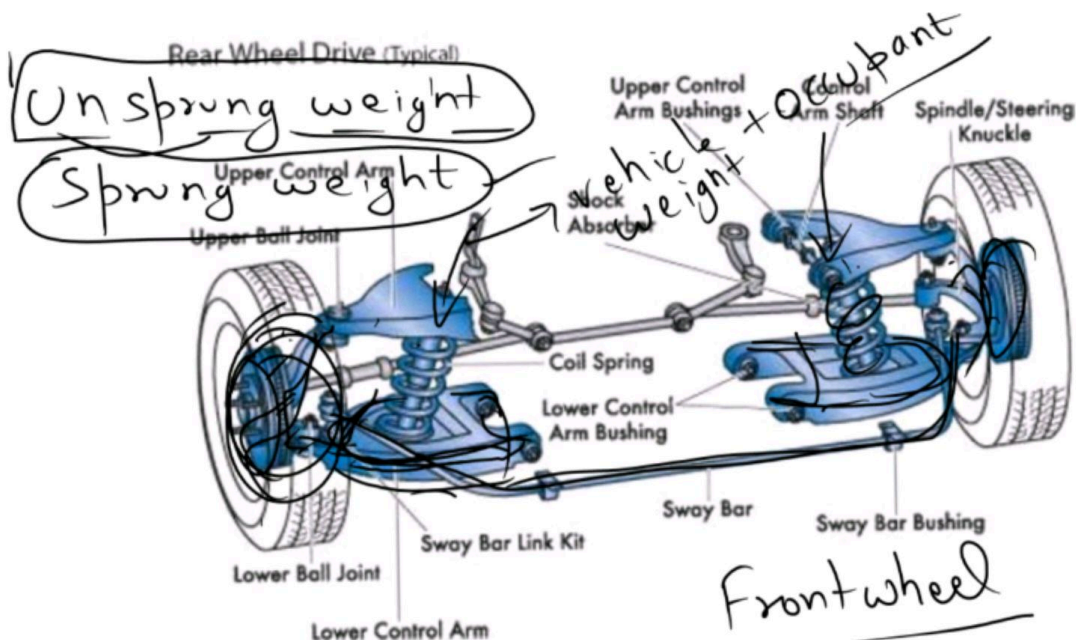
A suspension system is needed for following purpose:-

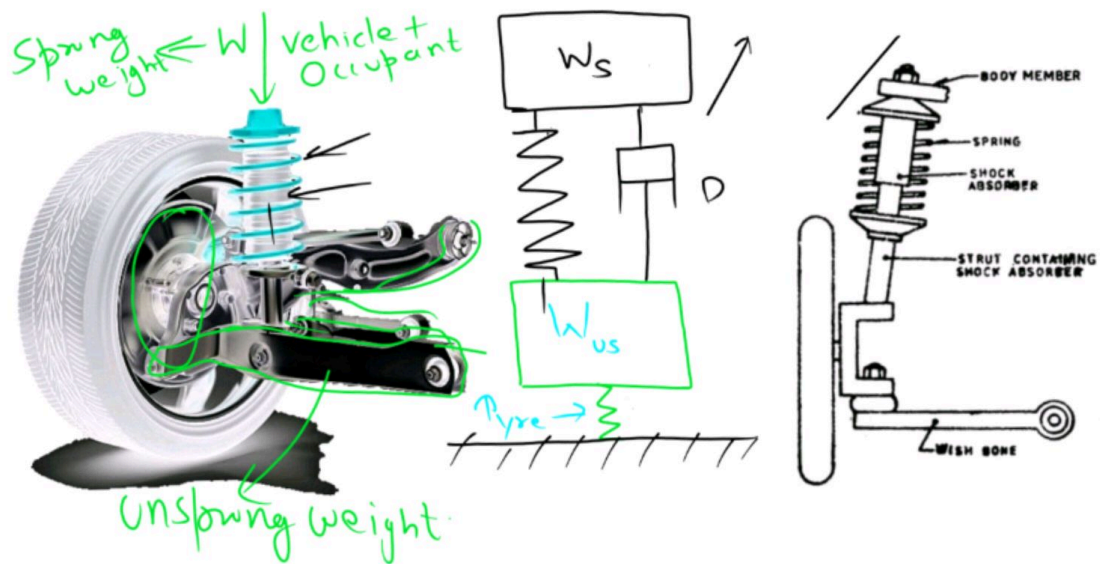
- To prevent the road shock being transmitted to the vehicle components.
- To protect the occupants from road shocks.
- To maintain stability of the vehicle in rough condition.



Elements of Suspension System

- A suspension system can be represented in a schematic form as shown in figure below where, W_{sp} and W_{us} are the sprung and unsprung weights respectively. R_s is the spring system, R_t is the tyre system and D is the damper, i.e. shock absorber.
- The sprung mass is the mass of the passenger carriage while the unsprung mass is the mass of the wheel axle system.



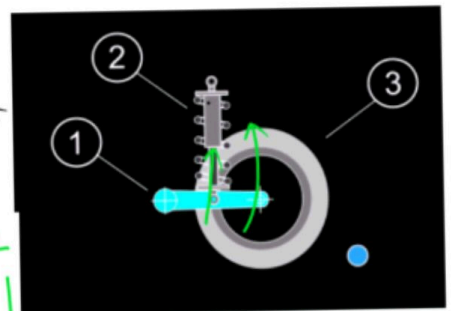
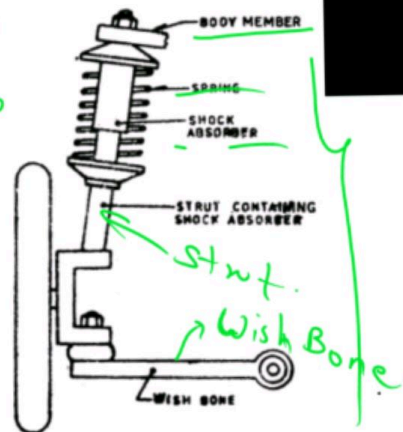


- Important elements of a suspension system are:

1. Springs → Shock & transmit
2. Dampers → shock absorber & 1

Working of suspension system

- when a vehicle encounters a bump, the tyre is suddenly pushed up, this force will compress the spring and did not allow the force to reach to the passenger unit. As the wheel passes over a bump, the spring expands at a much lower acceleration, and the vibrations are quickly killed by the damping action of the dampers.



Brake System

- A brake is a mechanical device that interrupt the motion by absorbing energy from a moving system. It is used for slowing or stopping a moving vehicle at will.

Need of Braking system/Function

- It is required to stop the vehicle at will with minimum distance in an emergency.

Types of Brake

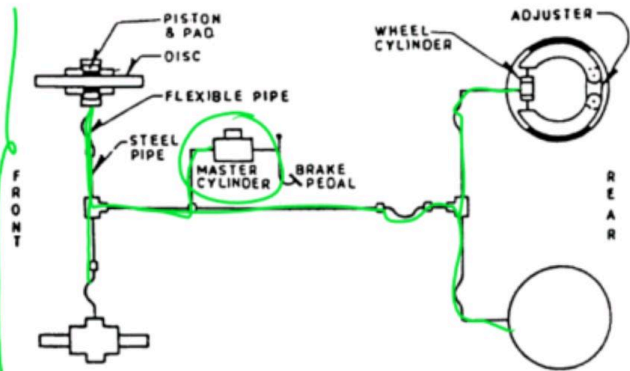
- There are various type of brakes used in automotive which are as follows

1. Mechanical Brakes
2. Hydraulic Brakes
3. Electrical Brakes
4. Vacuum Brakes
5. Air Brakes
6. By-wire brakes

4-wheeler → Automobile

Hydraulic Brakes

- Most of the car use hydraulically operated foot brakes on all the four wheel with an additional hand brake mechanically connected to the rear wheels. The front wheel is having disk brake which the rear wheel is having drum brake with one cylinder.
- The main Components of a hydraulic braking system is a master cylinder which contain reservior for the brake fluid and operated by brake pedal which is further connected to each wheel cylinder through steel pipe lines and flexible hoses.



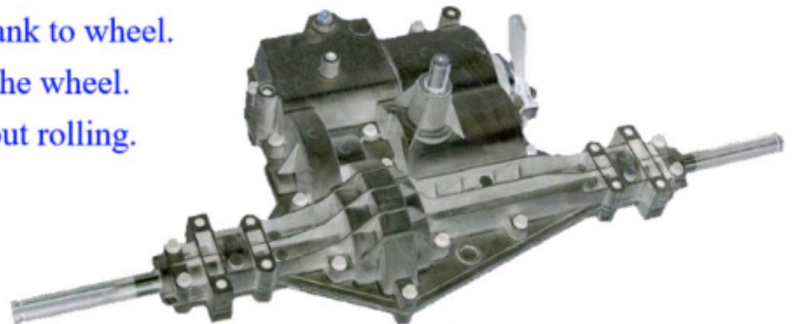
→ High pressure rubber tube

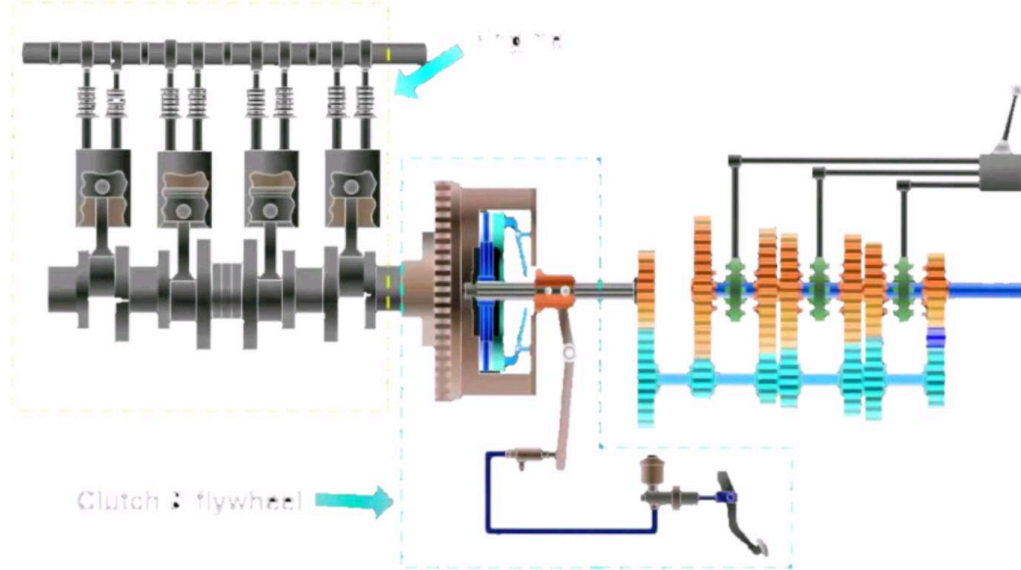
Transaxle

- A transaxle is a single mechanical device which combines the functions of an automobile's transmission, axle, and differential into one integrated assembly.

Function of transaxle

- It help to transmit power from engine crank to wheel.
- It help to achieve the required torque at the wheel.
- It help in proper turning of vehicle without rolling.





Noise Vibration and Harshness Control

- Noise, vibration, and harshness (NVH) is the study and measurement of aural (connected with hearing & listening) and tactile (sense of touch) feedback in an object.
- NVH analysis helps identify where a sound is coming from and why. Engineers can then reduce or eliminate unwanted noises, as well as enhance beneficial sounds.

There are three sources of NVH in vehicles:

1. Aerodynamics: HV/AC fans, Wind against the car body
2. Mechanical: Brake friction, Engine operations, Tire contact with the road
3. Electrical: Driver alerts, inverters in electric cars



There are three principal means of Controlling NVH:

- a. Reducing the source strength, as in making a noise source quieter with a muffler, or improving the balance of a rotating mechanism.
- b. Interrupting the noise or vibration path, with barriers (for noise) or isolators (for vibration).
- c. Absorption of the noise or vibration energy, as for example with foam noise absorbers, or tuned vibration dampers.

Body Balancing

- The term “Body balancing” refers to the distribution of the car’s weight across its four tires connecting with the road. If your vehicle is well-maintained, has good suspension and optimal tire pressure, it will be evenly balanced when stationary on a flat road.
- During running for maintaining perfect balance wheel balancing and rotating mass balancing must be done.
- Rotating mass balancing is done while design the engine and other rotating components while wheel balancing is done every time we change tyre.

Tyre Technology

- A tyre (or tire) is a circular and ring-like part of a vehicle that comes in contact with the ground.
- Tyres are fitted on rims and filled with compressed air. Natural rubber has been the most widely used material in the manufacturing of tyres. Modern tyres also employ materials like synthetic rubber, fabric, steel wires, carbon black.

Functions of a tyre

1. To maintain contact between vehicle and ground by providing the desired traction.
2. To support the load of the vehicle.
3. Dealing with various forces acting on the vehicle during its motion.
4. Providing cushions against shocks and damping them.

Tyre Designation

- Tyre is designated by alphanumeric tire code which is generally molded into the sidewall of the tire.
- The code specifies the dimension of tyre and some key limitation, such as load bearing ability and maximum speed.



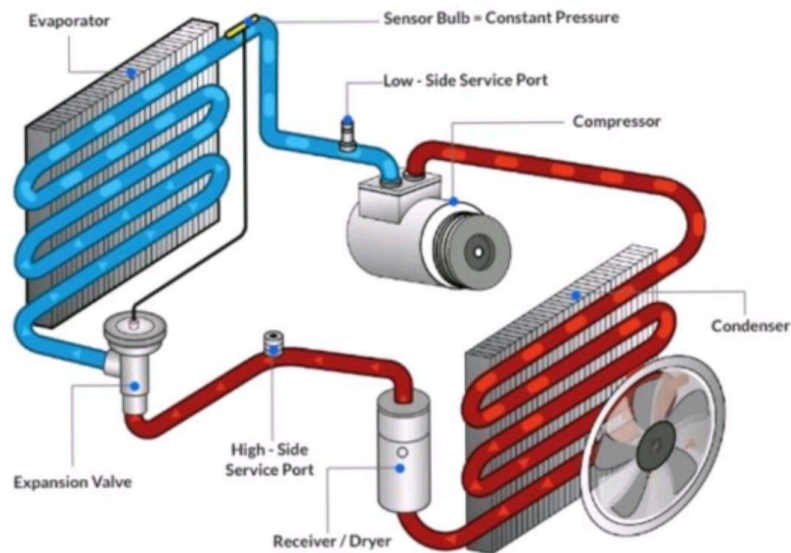
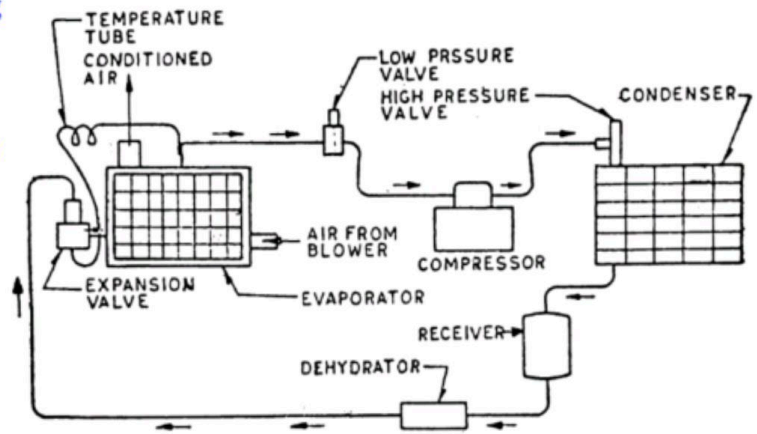
AC/DC Motor

- Motor is the power plant of the vehicle which provide motive power to run the vehicle.
- Motor may be AC or DC powered depends upon the requirement and types of control needed to operate the vehicle.

Air-conditioning and Heating System

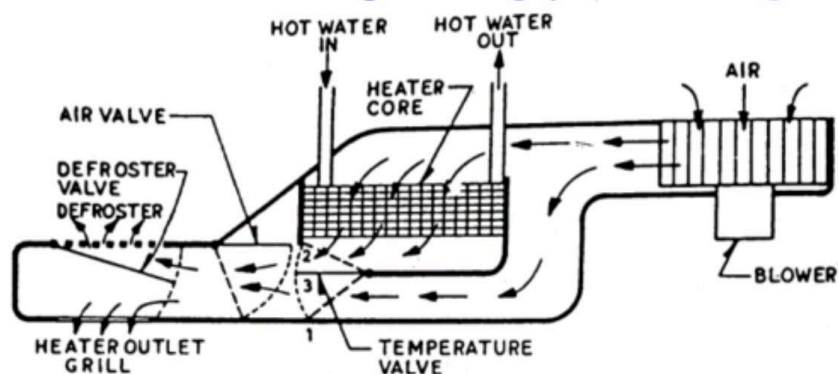
Air conditioning

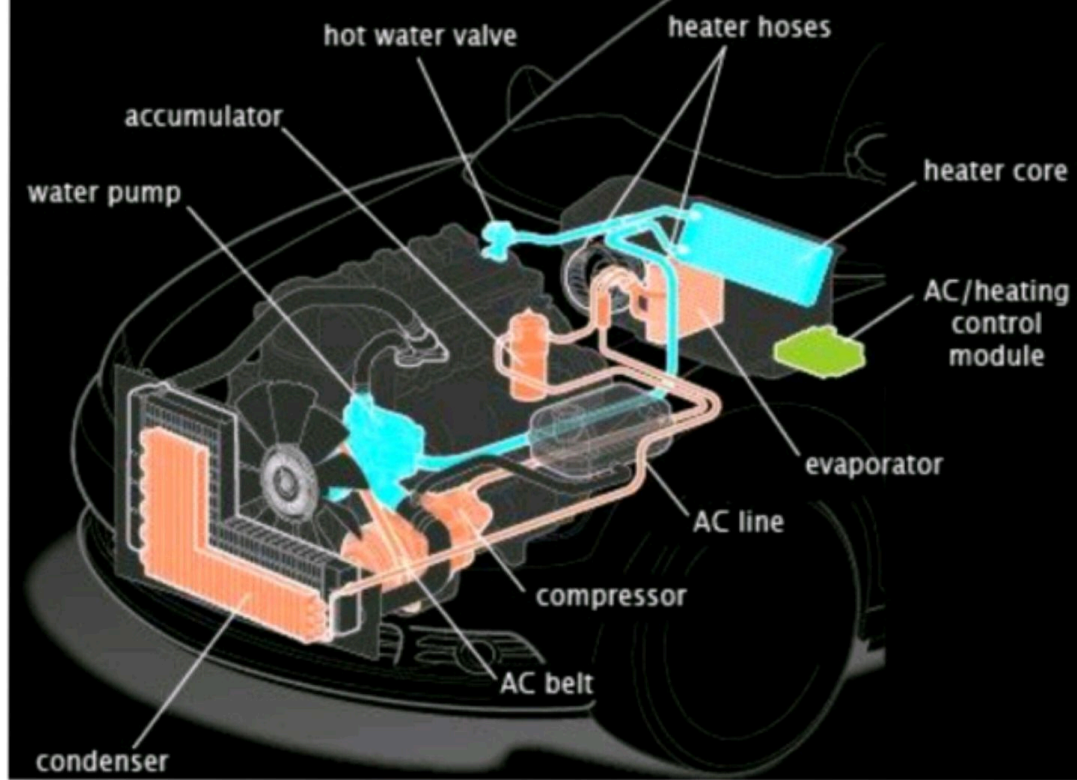
- The aim of automotive air conditioning is to control the temperature and humidity of the atmospheric air and circulate the same in the automobile so that a comfortable feeling is obtained by the passenger in summer season.
- The layout of the air conditioning system of car is shown in figure. It includes the compressor, condensor, evaporator, receiver-dehydrator and connecting lines.



Heating system

- The heating system in a vehicle is used to provide the desired air temperature inside the vehicle in winter season.
- The different sources of heat which can be used to rise the inside air temperature of vehicle are Hot air heater, Hot water heater & Petrol heater. Hot water heater is widely used to raise the inside temperature of the automobile.
- Hot water heater utilizes hot water from engine cooling system for heating the air.





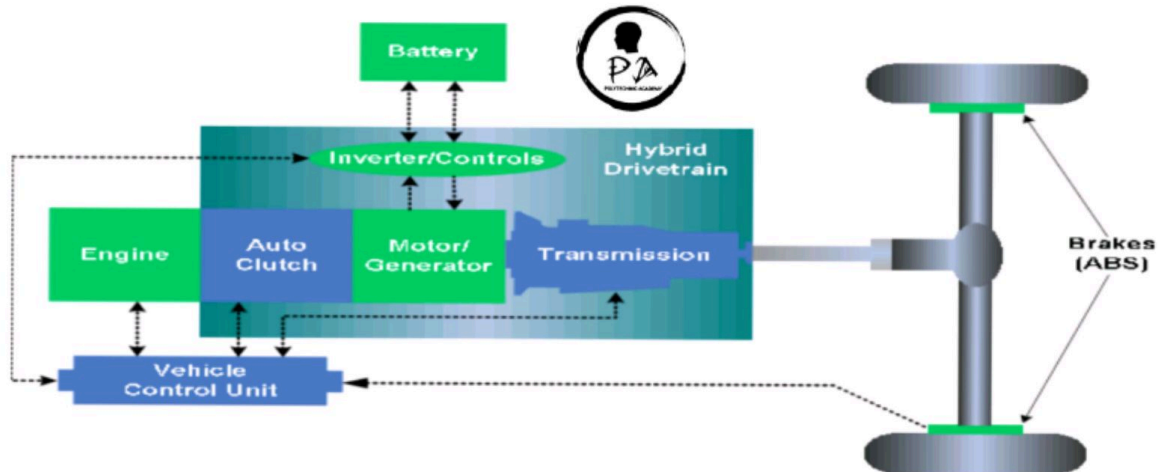
Lighting System

- The automobile lighting system consist of head light, side light, tail light , stop light and reverse light. A part from these there are instrumental panel light, direction indicator flash light and the light inside the body to light up passanger compartment.
- The purpose of head light is to help the driver to drive safely at reasonable speed at night or even during the day in low visibility condition.
- Side light are low intensity parking lamps.
- Tail lights are used to light up the back of the car so that the driver of the vehicle coming behind it will be able to see it. These are covered by red lenses so that they can be seen from distance.
- Stop light is illuminated when brake is applied it help in indicating the driver of the vehicle coming behind.
- Reverse light is illuminated when reverse gear is applied it help in indicating the driver of the vehicle coming behind that the vehicle in front of it is moving is back direction.



UNIT-3

EV POWER TRANSMISSION SYSTEM



Transmission System:-

- The mechanism that transmits the power developed by the motor to the driving wheels is called the transmission system (or Power Train).
- Basic types of transmissions can be Manual and Automatic Transmissions. But in electric vehicle only automatic transmission used.
- As the speed variations in electric vehicle is less and also automatic transmission gives comfort to driving the vehicle.
- Single speed Transmission and Multi-speed Transmission are two types of transmission mostly used in electric vehicles.
- The key factors affecting on the transmission are fuel efficiency, motor efficiency, speed torque variation, power, cost, weight etc.
- Based on the mentioned factors both transmission systems can be compared.

Types Of Transmission System :-

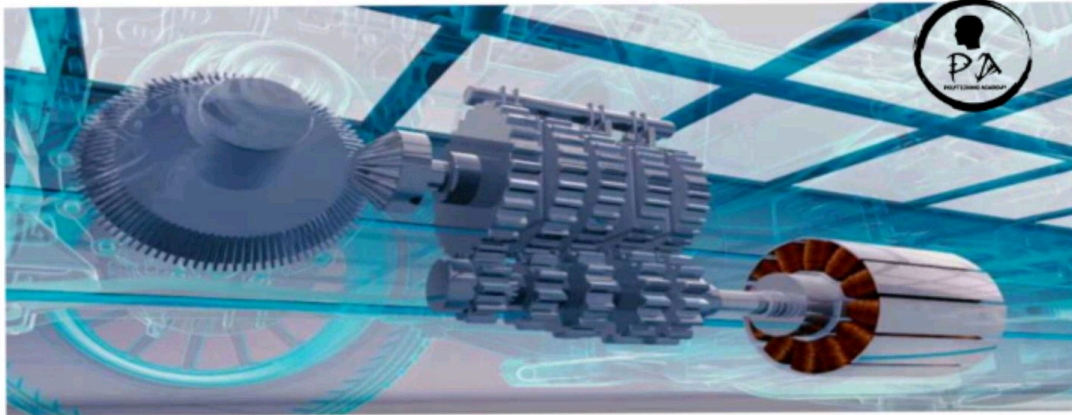
There are two types of transmission system

1. Single Speed Transmission
2. Multi Speed Transmission

1. Single Speed Transmission :-

- Most of the electric car are single - speed transmission vehicle.
- Speed - torque characteristics of electric machine make them most suitable for traction.
- The ability to meet high torque demands at low speed is the reason why a multispeed gearbox is unnecessary in an electric car.
- One of the biggest difference in the majority of EVs, when compare to conventional vehicles is the absence of a gear shifter, simply because they have just one gear.
- A diesel or petrol powered car's internal combustion engine develops its power and torque at different RPM's and therefore it needs a gearbox.
- EVs generally have very compact drive selector.
- After starting the motor, all the driver needs to do is push a button or turn a dial to select "D" to move forward, "R" to reverse and "p" to park.

- Some people may feel this is similar to what you do in a regular car with an automatic transmission or gearbox, but it is not.
- Automatic transmission have multiple gears and speeds that are selected and shifted automatically as per pressure on the accelerator or brake pedal.
- But most EVs have one gear, which means no gear shifts, either manual or automatic.



Advantages :-

1. It has direct drive.
2. Its operation is very smooth and also it generates instant torque when coupled with motor.
3. No complex gear system required.
4. It occupies less space.



Limitation :-

1. It has a fixed gear ratio.



2. Multi- Speed Transmission :-

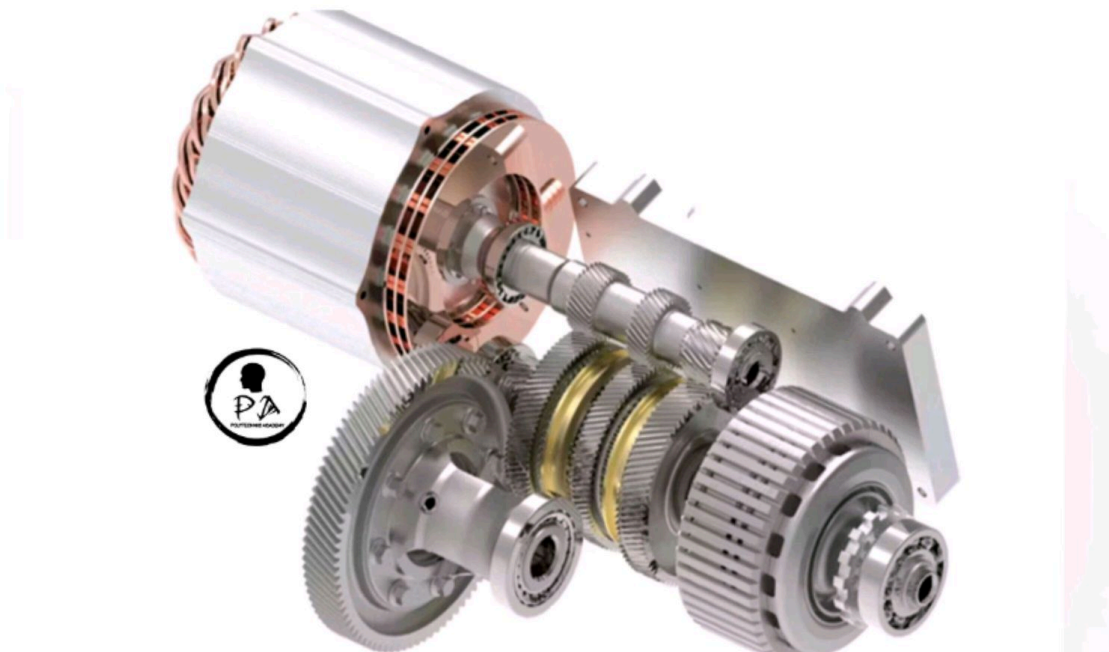
- In 99.99% EVs have single speed and no gear change arrangement.
- This does not mean, however, that a multi speed set up is not possible for an EV.
- With multiple gears, an EV could achieve better low end acceleration as well as top speed and this could also help increase its range marginally.
- In fact, some Formula E race cars have used gearboxes with up to five speed, while the Porsche Taycan employs a two-speed gearbox.
- The two speed transmission is only arriving now partly because it's complicated.
- Due to complexity of a multi-ratio gearbox as well as the added cost, all EVs come with no gearbox and have single speed unit.

Advantages

1. Stall torque can be increased.
2. Efficiency can be optimized.
3. Top speed of vehicle can be achieved.

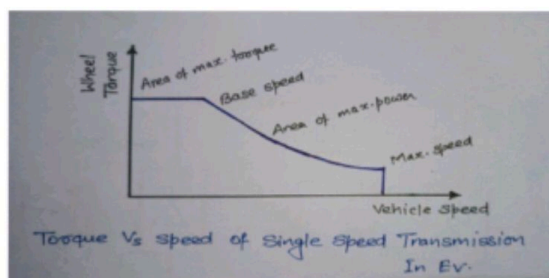
Limitations

1. These transmissions are too heavy and large.
2. It has very high cost.



Torque Vs Speed :-

For the comparison of two transmission systems based on torque vs. speed characteristics. We get following graphs which are based on experimentation on both of the transmission systems.



From above Fig 3 and Fig 4, we get torque vs speed characteristics for single speed and two speed gearboxes

EV Power Train

- Powertrain refers to the set of components that generate the power required to move the vehicle and deliver it to the wheels.
- As the name suggests, the powertrain provides power to the vehicle.
- Power train of an electric vehicle is a simpler system because it consists of very less number of components compare to the internal combustion engine.

Main Components Of An EV Powertrain :-

An EV powertrain has 60% fewer components than the powertrain of an ICE vehicle. The components are described below.

1. Battery Pack :-



- The battery pack is made up of multiple Lithium-ion cells and stores the electrical energy needed to run the vehicle.
- Battery packs provide direct current (DC) output.

2. DC-AC Converter :-

- The DC supplied by battery pack is converted to AC and supplied to the electric motor.
- This power transfer is managed by a sophisticated motor control mechanism (also referred to as Powertrain Electronic Control Unit) that controls the frequency and magnitude of the voltage supplied to the electric motor in order to manage the speed and acceleration as per driver's instructions communicated via acceleration/brakes.

3. Electric Motor :-

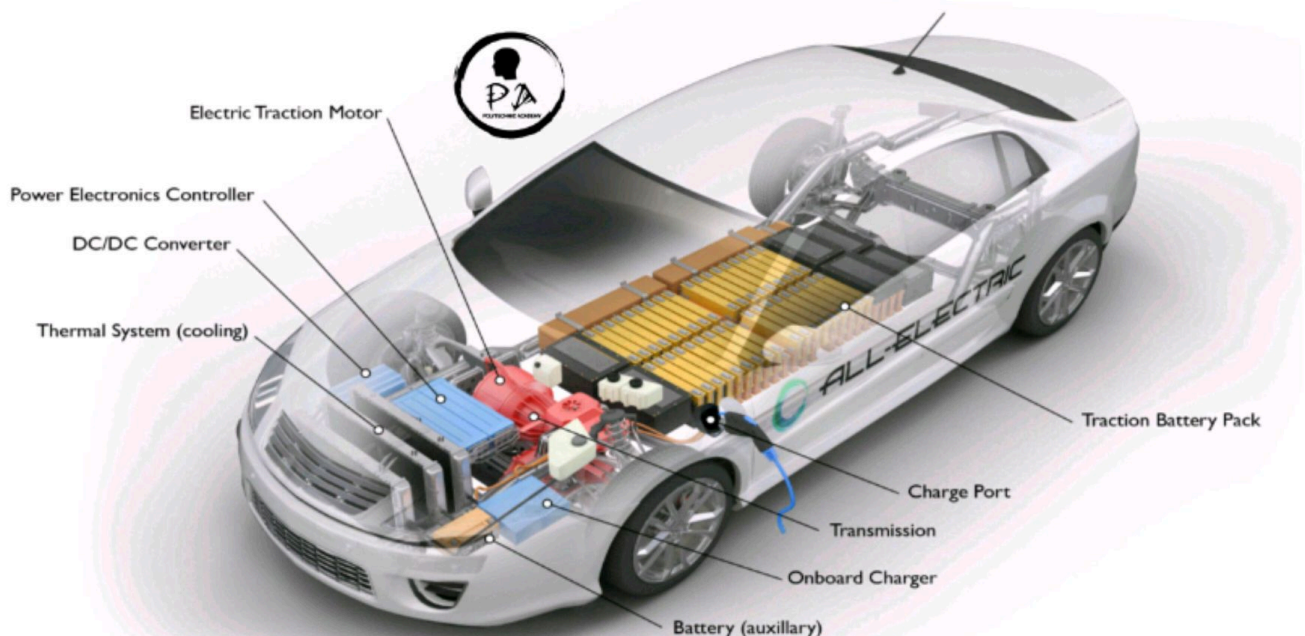
- Converts electrical energy to mechanical energy, that is delivered to the wheels via single ratio transmission.
- Many EVs use motor generators that can perform regeneration as well.

4. On-board Charger :-



- Convert AC received through charger port to DC and controls the amount of current flowing into the battery pack.

All-Electric Vehicle



Battery Parameters

Voltage :-

- The term “battery voltage” refers to the difference in electrical potential between the positive and negative terminals of a battery.
- A large potential difference results in a greater voltage.
- Electric potential is the difference in charge between two points – in this case, the two poles of a battery.
- One is positively charged, the other is negatively charged.
- A battery is also known as DC voltage source.



Current :-

- When a load is connected between both the positive and negative terminal of battery then electric current flow through the load.
- The direction of current is from positive terminal to negative terminal.
- The nature of current is DC (Direct Current).



Charging Rate (C-Rate) :-

- The charging rate is the rate of charging and discharging of the battery at which battery is full charge and discharge in one hour.
- The charging and discharging rate of battery is controlled by C- rate.
- The capacity of a battery is generally rated at the 1 C rate.
- For example, charging at a C-rate of 1C means that the battery is charged from 0-100% in one hours.
- A C-rate higher than 1C means a faster charge; for example, a 3C rate is three times faster, so a full charge in 20 minutes.
- Likewise, a lower C-rate means a slower charge: C/5 (or 0.2C) would be five times slower than 1C, amounting to a five-hour charge.

Battery efficiency

- Battery efficiency is define as the amount of energy you can get out of a battery relative to the amount of energy that's put into it.
- The amount of energy you get out is always going to be less than what's put in - however, there are specific factors that will affect that difference.
- During the charging and discharging one form of energy is converted into another form hence there is a power loss.
- The ideal value of battery efficiency is 80 to 90%.



$$\text{Battery efficiency} = \frac{\text{Energy output}}{\text{Energy supplied}}$$

Energy Density

- Energy Density is defined as the total amount of energy that can be store in battery per unit volume or per unit weight.
 - The energy density of battery is define in two parts
- a) Gravimetric :- Gravimetric is also known as specific energy and it is define as the capacity to stored energy per unit weight. It's unit is Whr/Kg.
- b) Volumetric :- It is defined as the capacity to stored energy per unit volume. It's unit is Whr/L.
- If gravimetry value of a battery is higher, then it stor more energy in less weight.
 - If volumetric value of a battery if higher, then it store more energy in small volume.

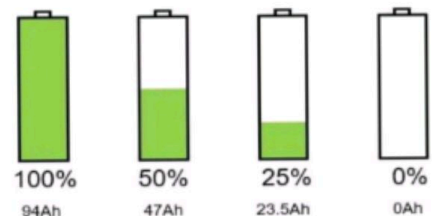


Power Density :-

- Power density is define as the capacity of energy deliver in a time period.
- If the value of power density of battery is high it means battery can supply more power in short time.
- Power density is expressed in W/Kg or W/L.

State Of Charge (SOC):-

- The state of charge (SOC) is a describe the amount of energy available in a battery.
- SOC expressed in %.
- 100% SOC means battery is full charged and 0 % SOC means battery is fully discharged.
- In an Electric vehicle, the range of vehicle is determine by the value of SOC.



State Of Health (SOH) :-

- The battery state of health is a measurement that indicates the level of degradation and remaining capacity of the battery.
- It is essentially the difference between the health of a new battery and the health of a used battery, and typically represented as a percentage of its initial capacity.
- SOH describe the remaining life of battery.
- SOH is expressed in %.
- Typically, a battery's SoH will be 100% at the time of manufacture and will decrease over time and use.
- The biggest factors that contribute to battery degradation are driver patterns, driver aggression, climate, cabin thermal dynamics, and infrastructure, with driver patterns and climate being the biggest.



Depth Of Discharge :-

- Depth of discharge (DoD) is a figure of merit that is often used instead of SoC.
- It is defined as an amount of charge removed from the battery at the given state (Qd) related to the total amount of charge, which can be stored in this battery (C) and usually expressed as a percentage:

$$\text{DOD} = \frac{\text{Removed amount of charge (Q}_d\text{)}}{\text{Maximum available amount of charge (C)}} \times 100$$



$$\text{DoD} = 100\% - \text{SoC}$$

Operating Temperature :-

- The operating temperature of a battery is the value of that temperature at which battery safely charging and discharging.
- The lithium ion battery should be charge at a range of 32 F to 113 F and discharge between -4 F to 131 F.
- Their charge and discharge performance remain good at a higher temperature compare to other batties.

- If lithiun ion battery expose in high temperature, then it's life span is shorter.

Life Cycle :-

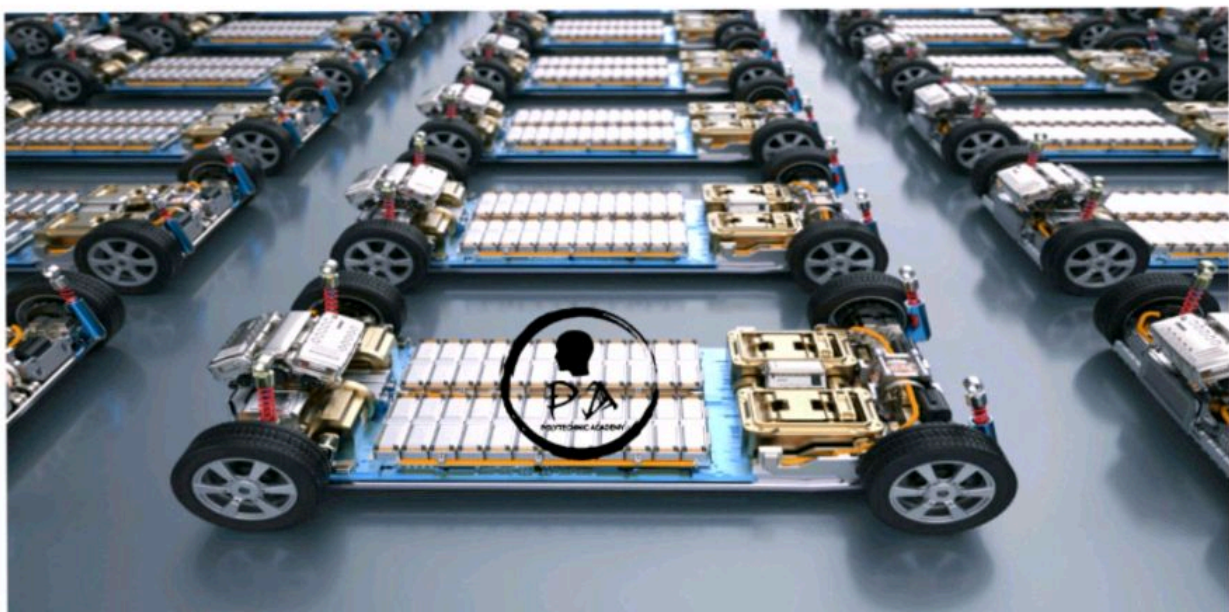
- The cycle life of batteries is the number of charge and discharge cycles that a battery can complete before losing performance.
- The cycle life of Li-ion batteries is affected significantly by the depth of discharge.
- Generally a Li-ion battery life cycle is 2000-3000 which is significantly longer than a lead acid battery which life cycle is 1000-1500.

Specific Energy → Gravimetry energy



Specific power → Power density

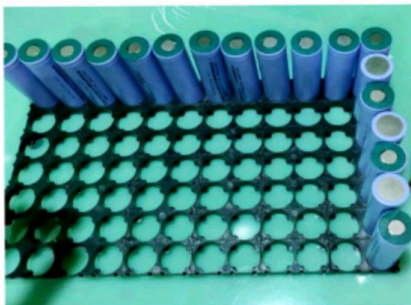
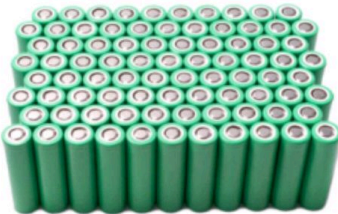
Battery Assembly



Battery Assembly

- Battery assembly is process in which number of cell (lithium Ion cell) are connected together to form a battery bank.
- For battery assembly following components are required :-

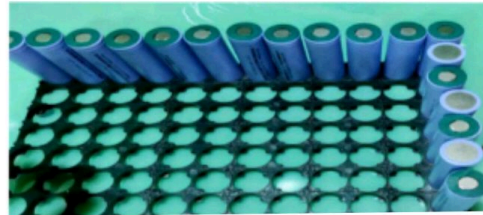
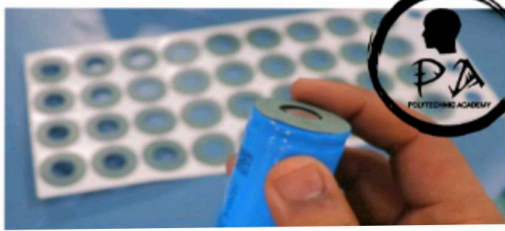
1. Lithium cell
2. Connector/Cell connector
3. Nickel strips
4. BMS (Battery Management System)
5. Different types of wire
6. Insulation paper.
7. Different types of tapes
8. Plastic/Metal container



Assembly Process :-

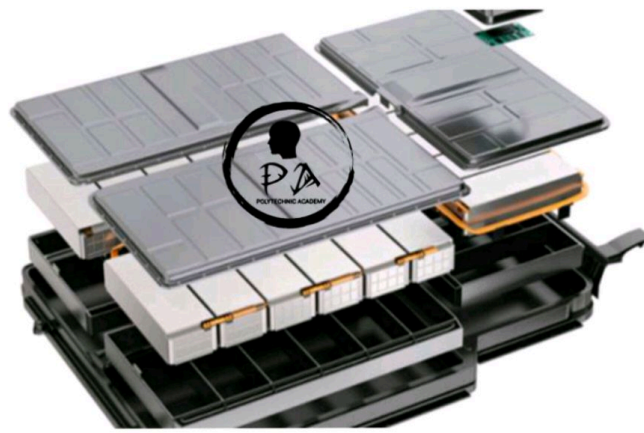
The assembly process describe below

1. Take the proper number of cell which are used in battery pack.
2. All cell must have the same voltage and current rating.
3. Stick barley paper around the positive terminal of battery and metal part of positive terminal remains open.
4. Barley paper is an insulating paper that is applied to the cell so that the battery would not have any internal short circuits. It can be done manually, but there is an automated machine to do that the "auto-insulating sticker pasting machine."
5. The next step is to arrange them together with the help of the cell holder. These are little square boxes connected to each other. You can combine as many cell holders as you want and put cells in them.



- It takes two layers of cell holders, one at the bottom and another at the top.
- In this, you can either connect cells in parallel or connect them by interchanging their terminals.
- Once your combination is ready, you have to solder them, but first they are checked visually to make sure the positive and negative terminals are set correctly.
- Because if there is even a single cell misplaced, then it could make the whole battery dysfunctional.
- Next the nickel strips are welded along the positive and negative terminal of cell by using an automatic spot welding machine.
- This can be done manually or with the help of a machine. This machine is called an automatic spot welding machine. Finally the battery pack is prepared.
- Now BMS is connected with each positive and negative terminal of cell.
- Once the welding is done, the battery gets put into a basic pack that is made of epoxy sheets, insulator paper and different types of tape.
- This is not the actual battery body, but this is how it is prepared before applying the BMS.
- The next step is to apply BMS, which is a battery management system.
- BMS is a computer circuit that is connected to the multiple string in the battery. It monitors the cells, their charging and discharging properties, their heating and other such issues.
- In short, BMS is known to be the brain of the battery.
- BMS optimizing, protecting and monitoring the battery's health.
- Now, these batteries are ready to be put in their bodies either in soft or hard bodies as per requirement.





Dismantling :-

- Dismantling is a process to separate each cell safely from the battery pack.
- Dismantling is needed when battery pack performance becomes very poor or dead.
- Dismantling is required before recycling of Lithium ion cell. The disposal of Lithium cell is very dangerous for our environment so its recycling is necessary.
- Following steps of the dismantling

1. Opening the casing :-

The protective case around the battery is unscrewed and adhesive and silicones are removed. There are often also electrical components and cables for liquid coolants (cooling) or air (ventilation) that have to be taken away.



2. Voltage measurements

These are taken before and during the dismantling process. The modules/cells are gradually removed from the battery pack in order to reduce the voltage for safe processing. Ultimately, all the cells are removed.



3. Temperature measurements

Besides the voltage, the temperature is also measured by means of a heat scan. If this rises above the ambient temperature, it is an indication that something is amiss. Cells that show no deviation from the ambient temperature can be safely stored.

4. After dismantling

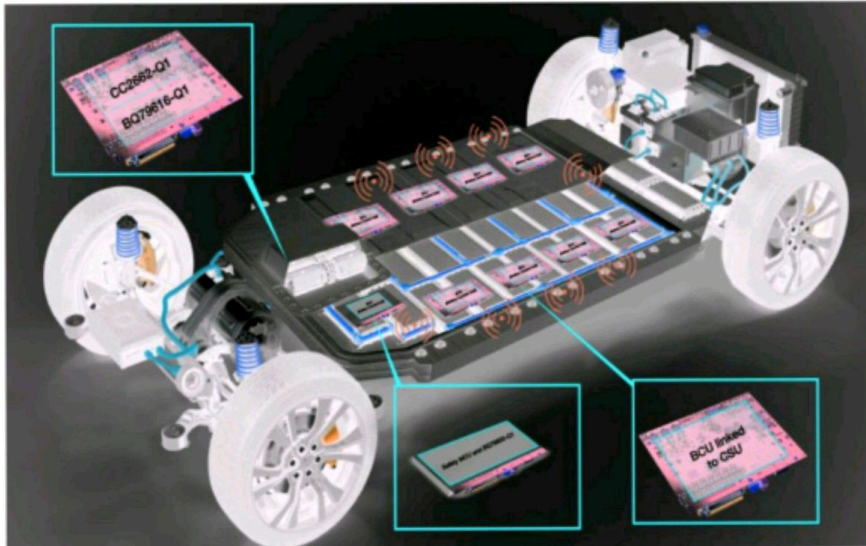
Once the battery has been dismantled, the battery cells are packed up again and - if necessary - stored in a specially secured area that is constantly monitored, among other things. From there they are taken to a recognised recycling company.

Safe Disposal Of Used Battery

Lithium-ion (Li-ion) batteries and devices containing these batteries should not go in household garbage or recycling bins. They can cause fires during transport or at landfills. Instead, Li-ion batteries should be taken to separate recycling or household hazardous waste collection points.



BATTERY MANAGEMENT SYSTEM(BMS)

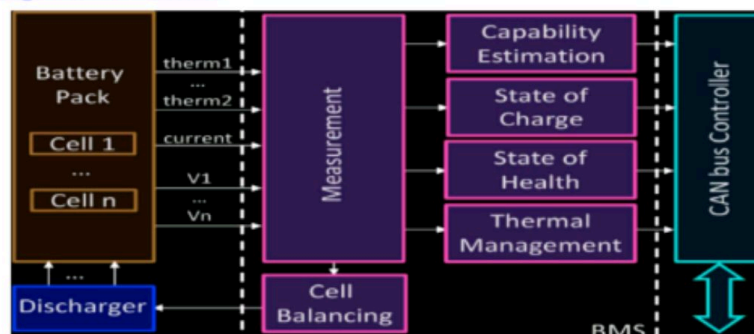


BATTERY MANAGEMENT SYSTEM(BMS)

- A battery management system is an electronic regulator that monitors and controls that charging and discharging of rechargeable batteries.
- Battery management system safely and efficiently operated the batteries in electric vehicle.
- Generally BMS used for battery pack of lithium ion cell.
- Lithium ion cells required more care during charging and discharging hence it required protection against overcharging and discharging.
- In electric vehicles mostly lithium ion battery packs are used, hence in that electric vehicle BMS must used.



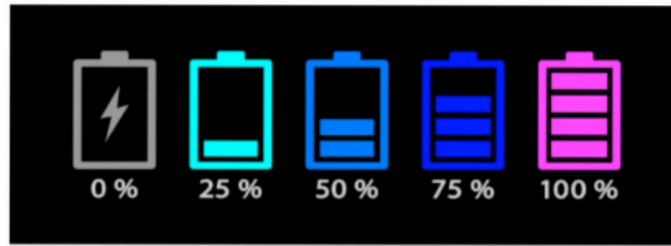
Block Diagram of BMS



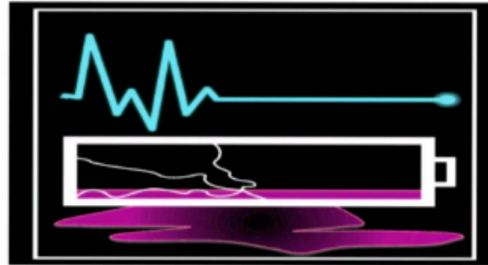
There are following functional block of BMS

1. **Battery Pack**:- Battery pack consists of many number of cell those are charging and discharging together.

2. **State of Charge(SOC)** :- SOC indicates the power remaining in battery.

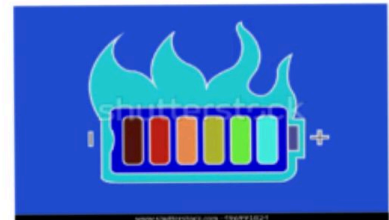
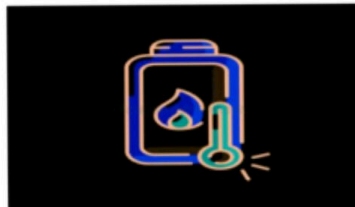
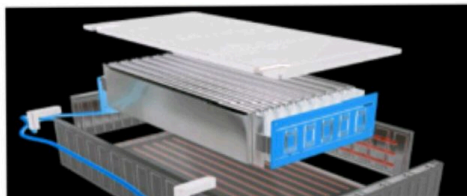


2. State of Health(SOH) :- The battery state of health is a measurement that indicates the level of degradation and remaining capacity of the battery. It is essentially the difference between the health of a new battery and the health of a used battery, and typically represented as a percentage of its initial capacity.



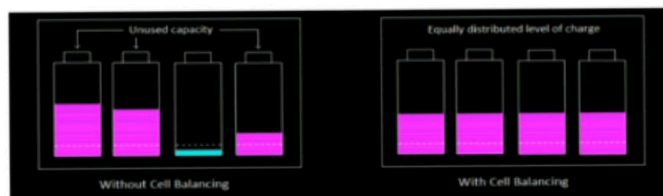
3. Thermal Management

Thermal management system is the part of BMS which is responsible for the managing/dissipating the heat generated during the electrochemical process. It allowing the battery to operate safely and efficiently.



4. Cell Balancing

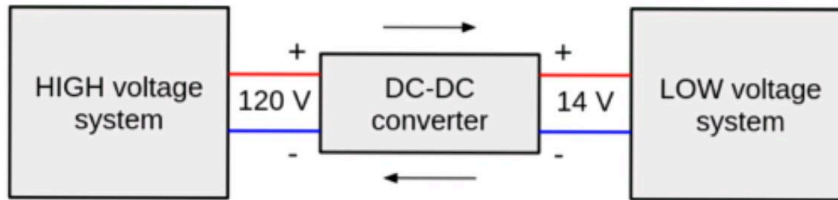
Cell balancing is the process of equalizing the voltages among individual cells.



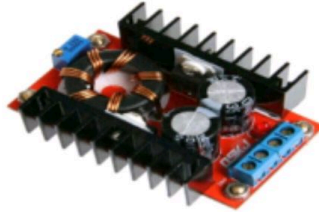
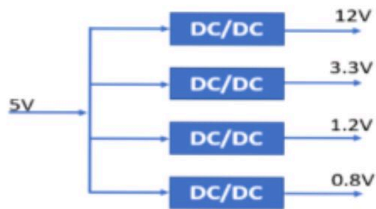
Advantages of Battery Management System:

1. Ensure that the battery is in good working order.
2. Battery health is continuously monitored to avoid an explosion.
3. Extends the battery's life expectancy.
4. Displays the battery level.

DC TO DC CONVERTER



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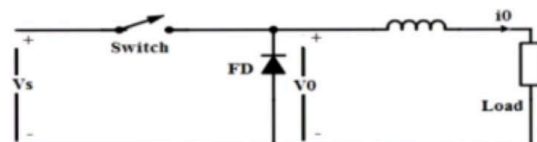


DC TO DC CONVERTER

- The DC-to-DC converters convert one level of DC voltage to another level.
- A DC to DC converter full fill the requirement of different values of voltage demand by electronics devices in electric vehicle.
- In power electronic , the DC to DC converter is known chopper circuit.
- The chopper is an automatic ON/OFF switch.

Circuit Diagram

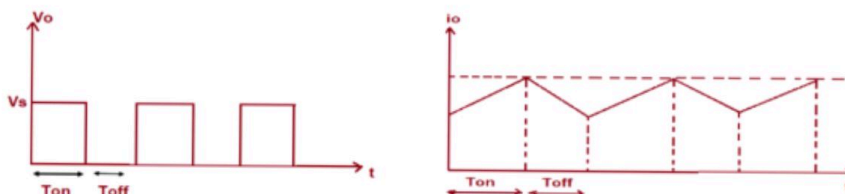
- Its circuit is designed by using a switch(chopper),a freewheel diode and an inductor.
- The automatic switch is connected with input voltage source and freewheel diode is connected in parallel with load.
- An inductor is connected in series with load.
- The output voltage is obtained across the freewheel diode.



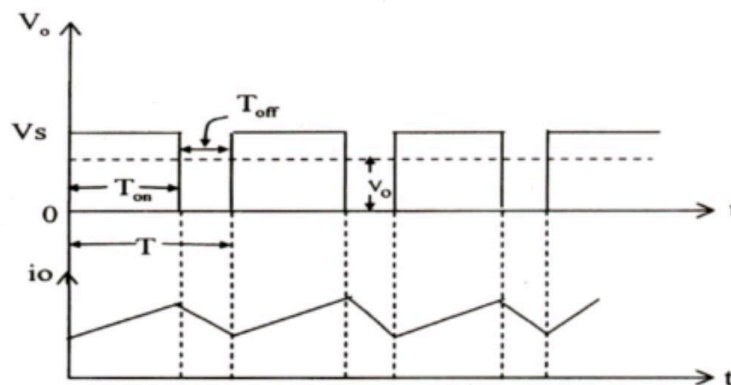
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Working

- During the period T_{ON} , chopper is ON and load voltage is equal to source voltage V_s . The inductor is charging and load current exponentially increasing across the load. During T_{ON} freewheel diode is OFF.
- During the periode T_{OFF} , chopper is OFF and the load voltage is zero because of freewheel diode is forward bias and work as short circuit. During T_{OFF} the inductor supply the current at load and the current is in decreasing order.
- The above process is repeated for each T_{ON} and T_{OFF} periode.
- During the working of this device load current contineously flow through the load.

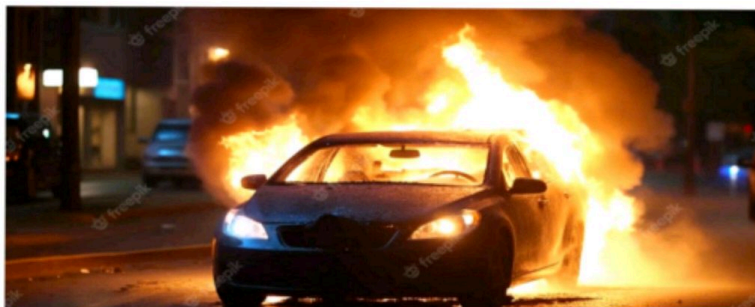


Output waveform



Thermal Management System & Body Control Module

- To ensure battery safety and extend battery life, an efficient battery thermal management system is required.
- During the charging and discharging process of the battery, a large amount of heat is generated to raise the temperature.
- Excessive temperature may cause the battery to escape gas or corrode.



There are following thermal management system

1. Air based battery thermal management system

- In this thermal management system let air to pass through the battery pack.
- Air enter in the battery pack from one side and exist from the other side. So heat of battery pack come out with the air.
- According to the flow of air outside the battery pack, it can be divided into two types of heat exchange methods: natural convection and forced convection.
- In natural convection method vehical is not specially designed but in force convention method vehicle is specially design to passes the air from one side to anoher side.

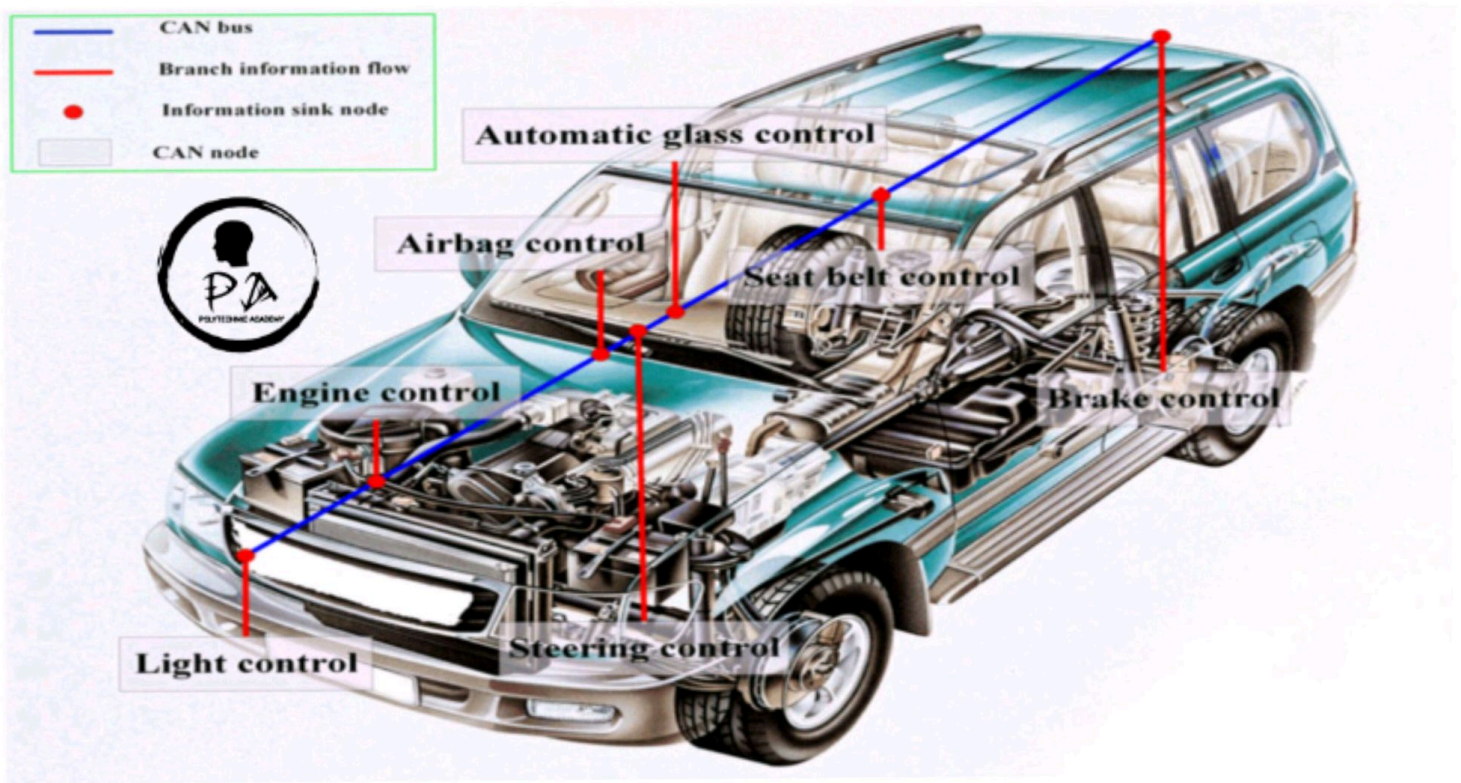
2. Air cooling systems

- For air cooling systems, it can be combined with the design of the vehicle's driving characteristics.
- The heat is taken away according to the natural wind formed by the speed of the vehicle.
- The battery can also be cooled by forced air circulation through fans and fans.
- According to the arrangement of the batteries, it can be divided into two types of ventilation: serial and parallel.

3. Liquid based battery thermal management system

- In this thermal management system liquid (coolant oil) is used for manage the battery heat.
- The pipe line is place in the battery pack which is filled with either water, ethylene glycol or refrigerant.
- The liquired flow through the pipe line and absorbe the heat of battery.
- The heated oil exchange our heat with natural air.
- In some cases battery pack directly immersed in the liquid.
- This method is costly than the above two methods.

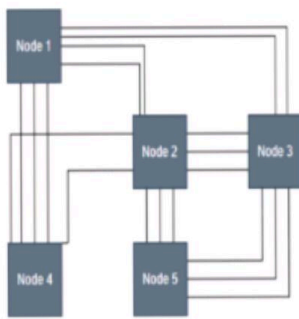
CONTROLLER AREA NETWORK (CAN)



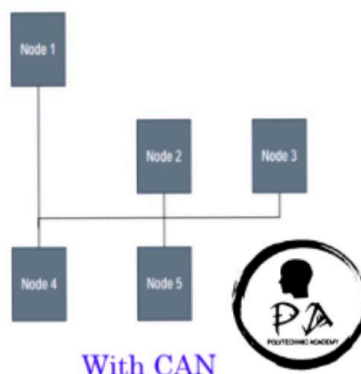
CONTROLLER AREA NETWORK (CAN)

- A Controller Area Network (CAN) bus is a communication system made for vehicle intercommunication.
- This bus allows many microcontrollers and different types of devices to communicate with each other in real time and also without a host computer.
- CAN is similar to nervous system of human body, which communicate all part of the body.
- This provides the nodes with information regarding the priority and the urgency of the transmitted message.
- It is a completely message-based protocol, and is used mainly in vehicles.
- A modern car can have upto 70 ECUs.
- In EV CAN allow ECUs to communicate with each other without complex wiring in between them.
- It consists of two electrical wires called CAN Low and CAN High.
- The information within each vehicle is transmitted from and to ECUs.
- It is used broadly in electric vehicle as well as trucks, buses, industry vehicles, ships, planes etc.

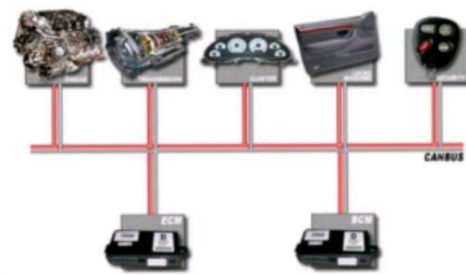




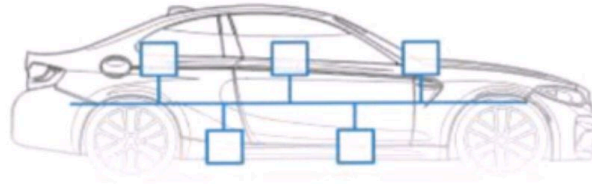
Without CAN



With CAN

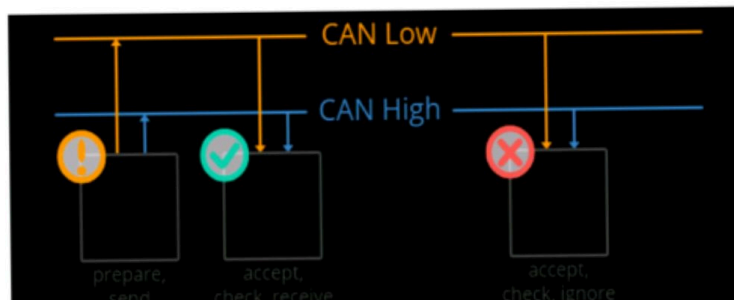


Controller Area Network (CAN)



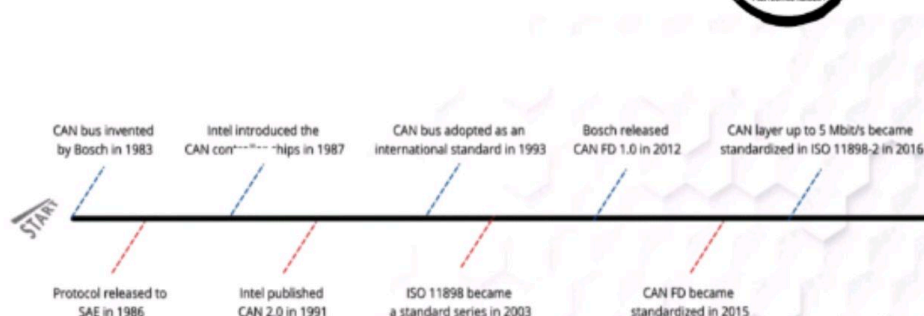
Working of CAN

The CAN bus system enables each ECU to communicate with all other ECUs - without complex dedicated wiring. Specifically, an ECU can prepare and broadcast information (e.g. sensor data) via the CAN bus (consisting of two wires, CAN low and CAN high). The broadcasted data is accepted by all other ECUs on the CAN network - and each ECU can then check the data and decide whether to receive or ignore it.



History Of CAN Bus

- Bosch originally developed CAN in 1985 for in-vehicle networks.
- In the past, automotive manufacturers connected electronic devices in vehicle using point-to-point wiring systems.
- Manufacturer began using more and more electronics in vehicles, which resulted in bulky wire harnesses that were heavy and expensive.
- CAN replaced bulky wiring system.
- It reduce wiring cost, complexity and weight.



Key Feature of CAN

The five key features of CAN are following

1. Low Cost :

ECUs communicate via a single CAN interface which reduce cost and weight.

2. Robust :

The system is robust towards failure of subsystems and electromagnetic interference making it ideal for vehicle.

3. Centralized :

The CAN bus system allows for central error diagnosis and configuration across all ECUs.

4. Efficient :

CAN message and prioritized based on IDs so that the highest priority IDs are non-interrupted-critical in vehicle.

5. Flexible :

Each ECU contains a chip allowing it to receive all transmitted messages, decide relevance and act accordingly. It allows easy modification and additional nodes.



UNIT-5

EV CHARGING TECHNOLOGIES

Charging Technologies :-

- Charging technologies is a technique which is used to charge the electric vehicle battery.
- Nowadays different types of charging technologies used for charging the electric vehicle battery.

Classification Of Charging Technologies :-

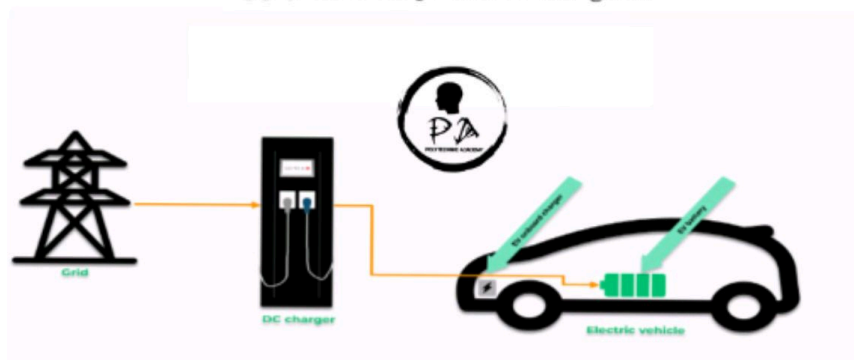


According to direction of flow of electric power there are following types of charging technologies

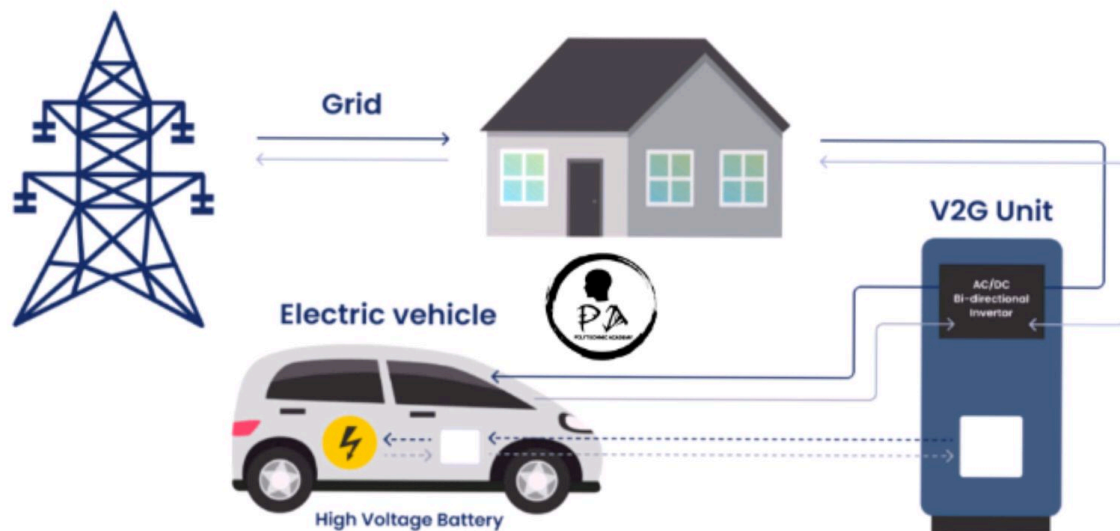
1. Grid-to-Vehicle (G2V)
2. Vehicle-to-Grid (V2G)
3. Bi-directional EV charging system

1. Grid-to-Vehicle (G2V) :-

- In Grid to Vehicle (G2V) charging system the electric power is supplied to the EV from the grid.
- This is the most common used technologies.
- It is an unidirectional charging system, it means electric power flows only from grid to vehicle.
- In this case vehicle can not supply electric power to the grid.



2. Vehicle-to- Grid (V2G) :-



- Vehicle-to-grid or V2G is a technology that enables energy to be pushed back to the power grid from the battery of an electric vehicle (EV).
- With V2G technology, an EV battery can be discharged based on energy production or consumption nearby.
- It is also known as V2X which means Vehicle to Grid (V2G), Vehicle to Building (V2B) and Vehicle to Home (V2H).
- V2G technology powers bi-directional charging, which makes it possible to charge the EV battery and take the energy stored in the car's battery and push it back to the power grid.
- It is a bidirectional charging system, it means electric power flows from power grid to EV and EV to power grid.
- This is the best way to utilize the renewable energy.
- The solar and wind energy is the renewable energy source but they are not able to supply continuous energy. The best way to utilize such type of renewable energy is V2G charging.

Why Vehicle to Grid Make Sense

- An average personal vehicle is on the road only 4–5% of the day, which means that a great majority of the day the vehicles are parked.
- At least 90% of personal vehicles are parked even during peak traffic hours.

Benefits of V2G Vehicle Own

- Consumers may profit from the use of electric vehicles (Evs) because electricity is cheaper than petrol for equivalent distances traveled.
- EVs in a V2G configuration could provide additional revenue to owners who wish to sell power back to the grid.
- EV owner charge the electric vehicle in low load time (night) and electric power supply back to grid at the time of full load or maximum demand.

Benefits of V2G Vehicle

There are following benefit of V2G vehicle

1. Renewable energy utilizes in proper manner.
2. Dependency on non-renewable energy decreases.
3. Environment get less polluted.
4. EV owner can earn profite.
5. Network operators can optimise investments & stabilise the grid



How Does V2G Work?

- When it comes to using V2G in practice, the most important thing is to make sure that EV drivers have enough energy in their car batteries when they need it.
- For example, a driver must be able to make a trip to work and back, at any time.
- This is the basic requirement of V2G and any other charging technology: The EV driver must be able to communicate when they want to unplug the car and how full the battery should be at that time.
- With Virta's V2G solution, the car battery is always charged to 70-90% when the driver needs to go.



3. Bi-directional EV charging system

- Bidirectional charging allows energy to flow both ways – in and out of your car.



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